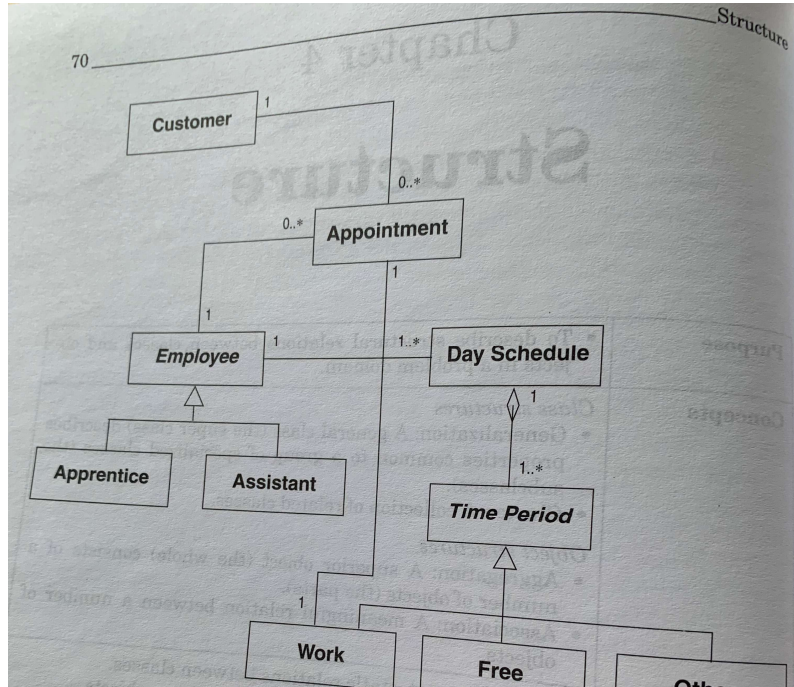




System Development

Session 2 - Problem Domain: Classes, Events and Structure

Anders Bruun



Classes Activity

The Event Table

50

<i>Events</i>	<i>Classes</i>				
	Customer	Assistant	Apprentice	Appointment	Pla
reserved	✓	✓		✓	✓
cancelled	✓	✓		✓	
treated	✓			✓	
employed		✓	✓		
resigned		✓	✓		
graduated			✓		
agreed		✓	✓		✓

Figure 3.1: Event table for the Hair Salon System (Chapter 20)

3.1 Classification of Objects and Events

Classification is a daily human activity. Children use words to describe their experiences. Adults create order in their activities by classifying phenomena according to certain categories. We all use classification to understand and structure reality, and to communicate with others. Although many classifications are part of our daily life, we sometimes use classification as a more conscious means of structuring something or creating order. This is especially true for certain professionals, such as librarians, lawyers, and system developers. Librarians, for example, carefully structure libraries to make books and information easy to find.

In the class activity, we focus on objects of the users' work with the goal of creating and selecting relevant abstractions. We emphasize this in the first of three class activity principles:

Principle: Classify objects in the problem.

Classes Activity

Selecting elements of a Problem Domain model

Object (look for nouns)

An entity with identity, behaviour and state

Class (look for nouns)

Collection of objects sharing structure, behavioural pattern and attributes

Event (look for verbs)

Instantaneous incident involving one or more objects

Events Classes

	Student	Lecturer	Course	...
Signed up	x			
Employed		x		
Exam planned		x	x	
Created			x	
...				

Affirmation Criteria

Class

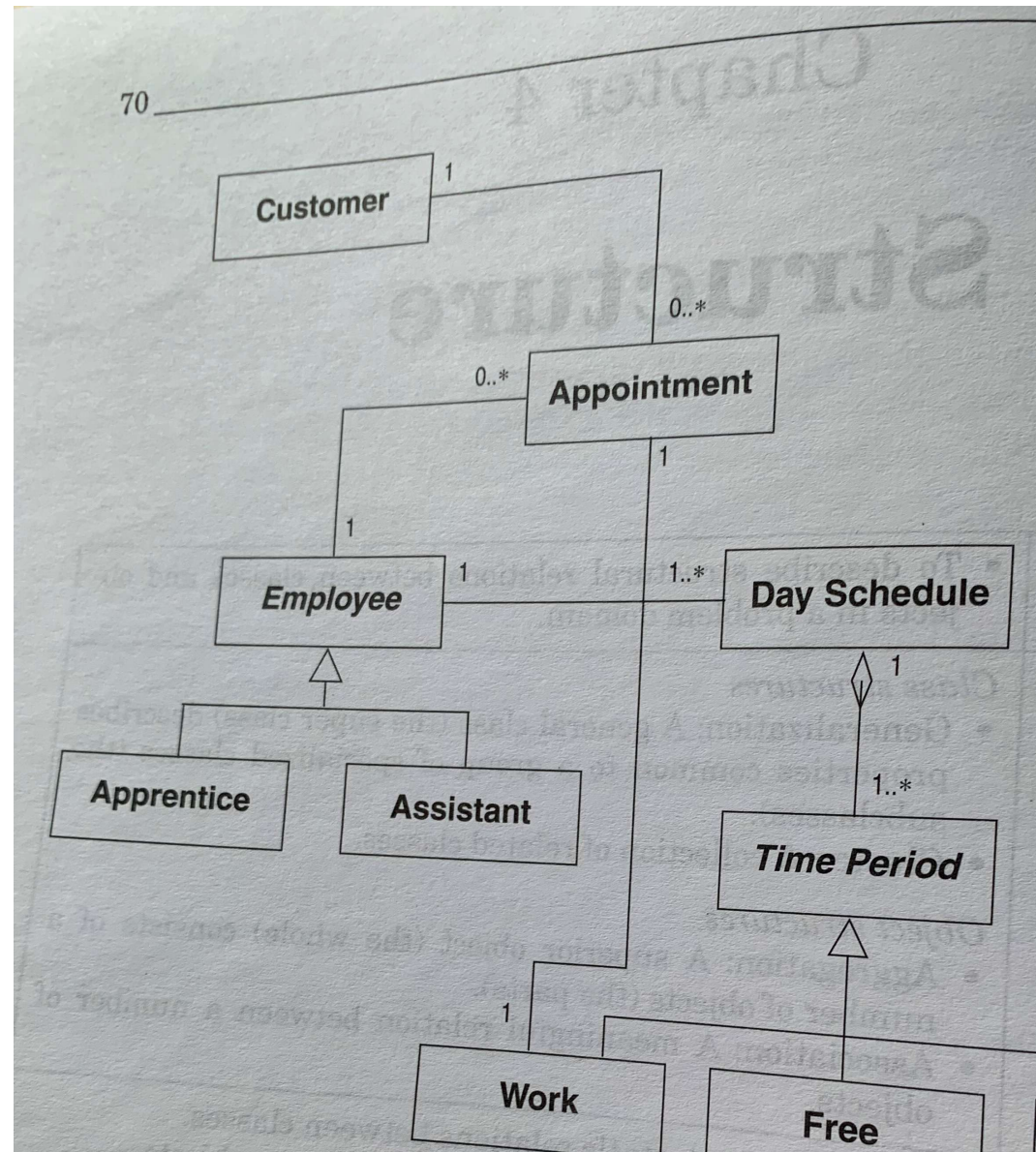
Can you identify objects?
Is it unique?
Multiple objects?
Suitable number of events?

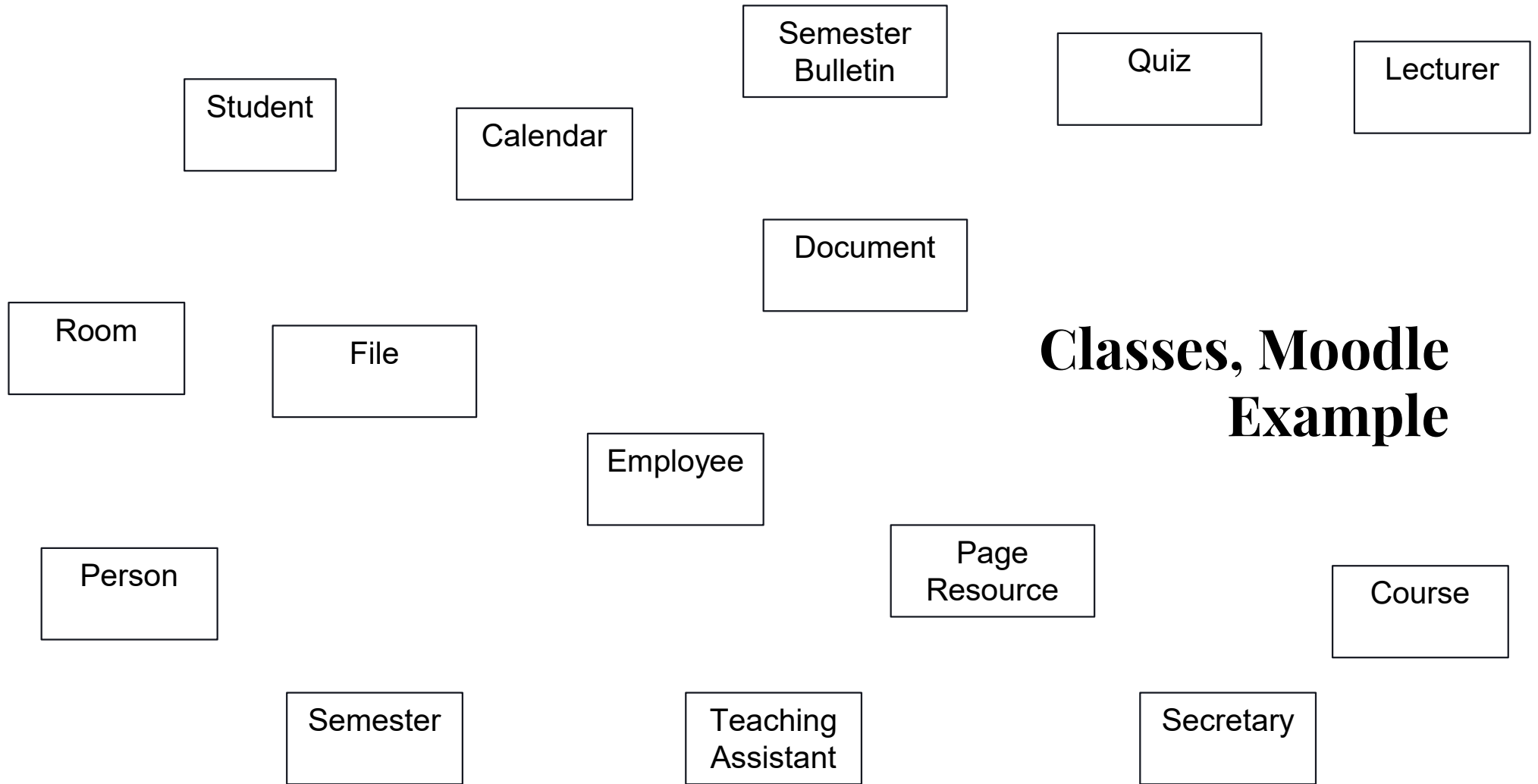
Event

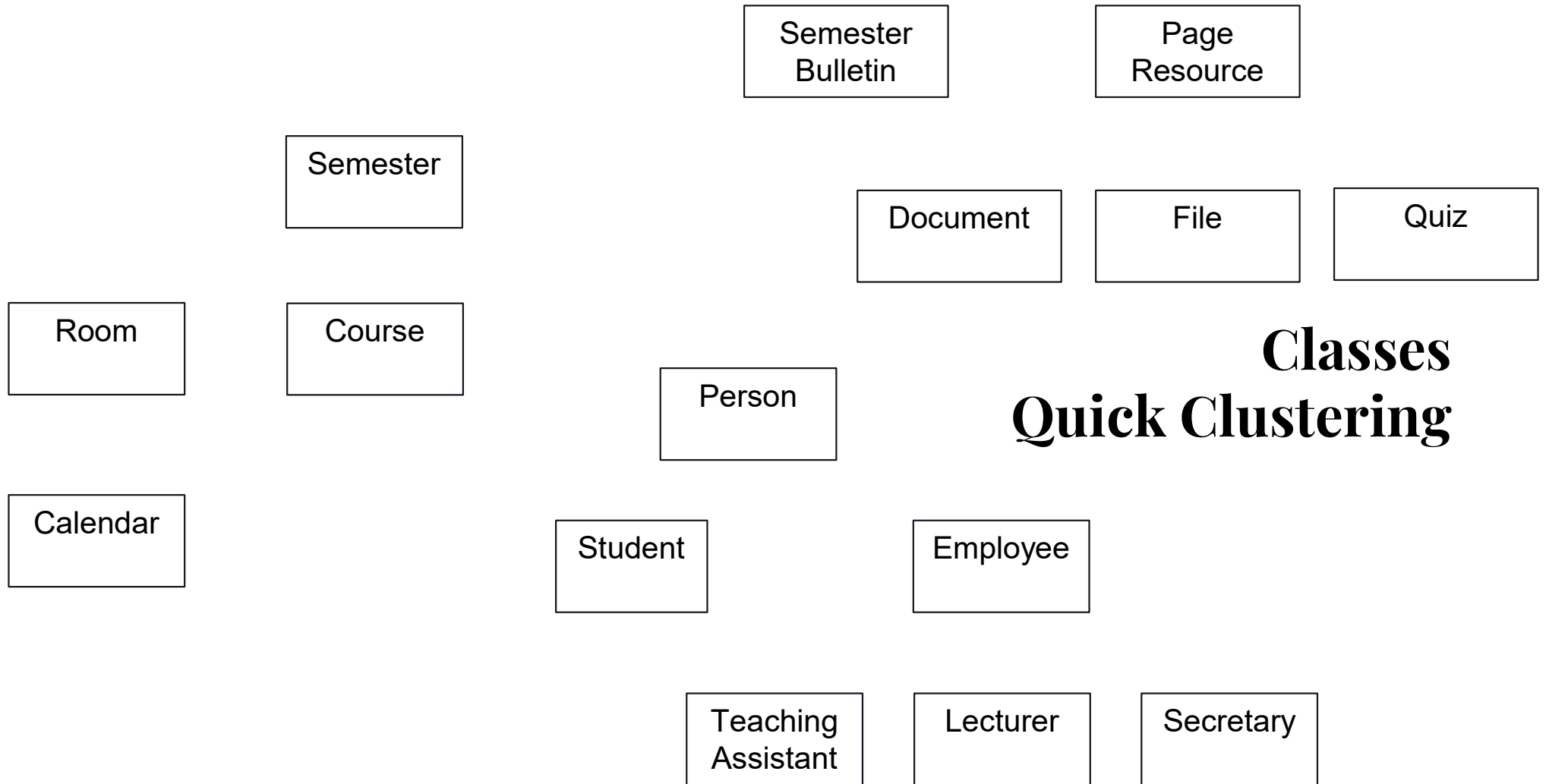
Instantaneous?
Atomic?
Identifiable when it happens?

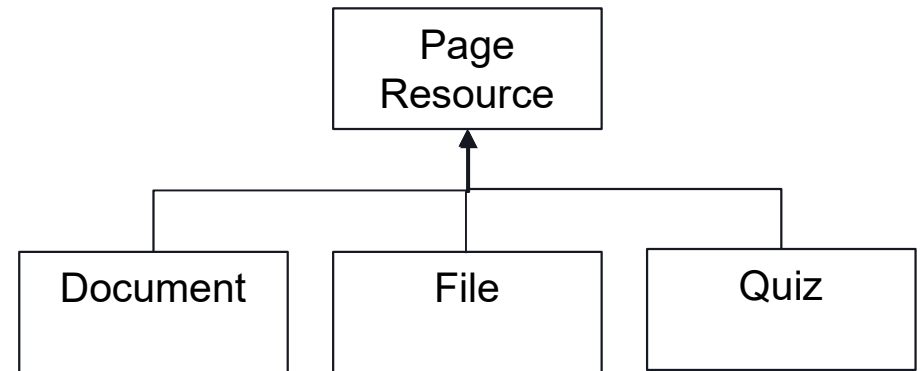
Structure Activity

The Class Diagram



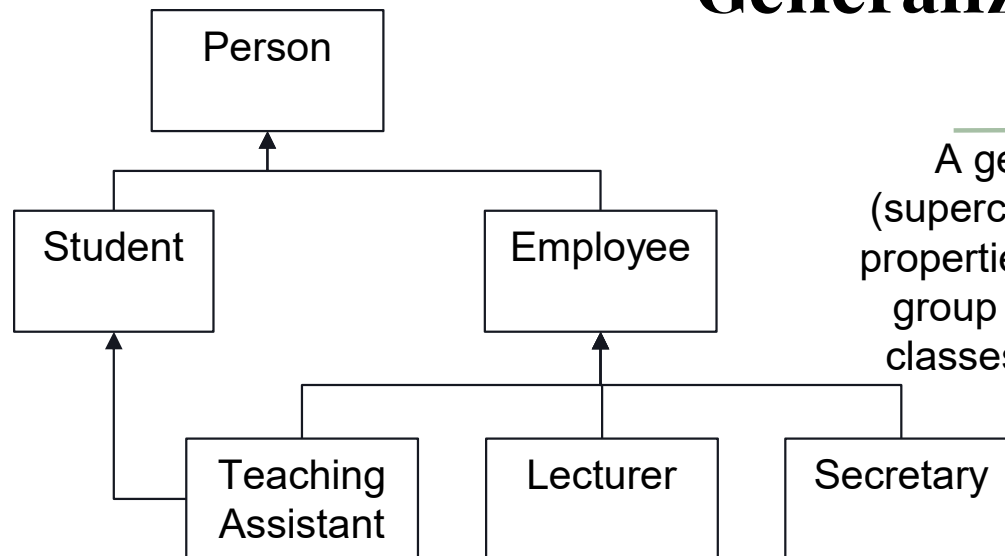




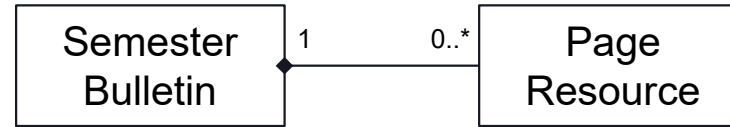
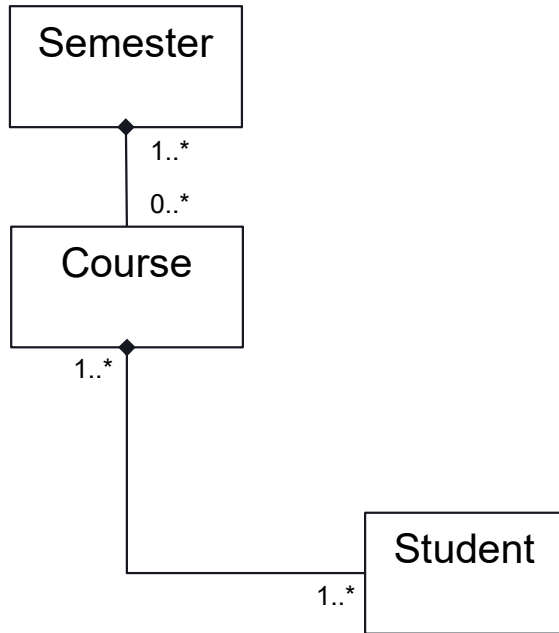
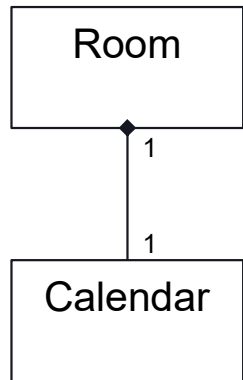


Generalization

Is-A



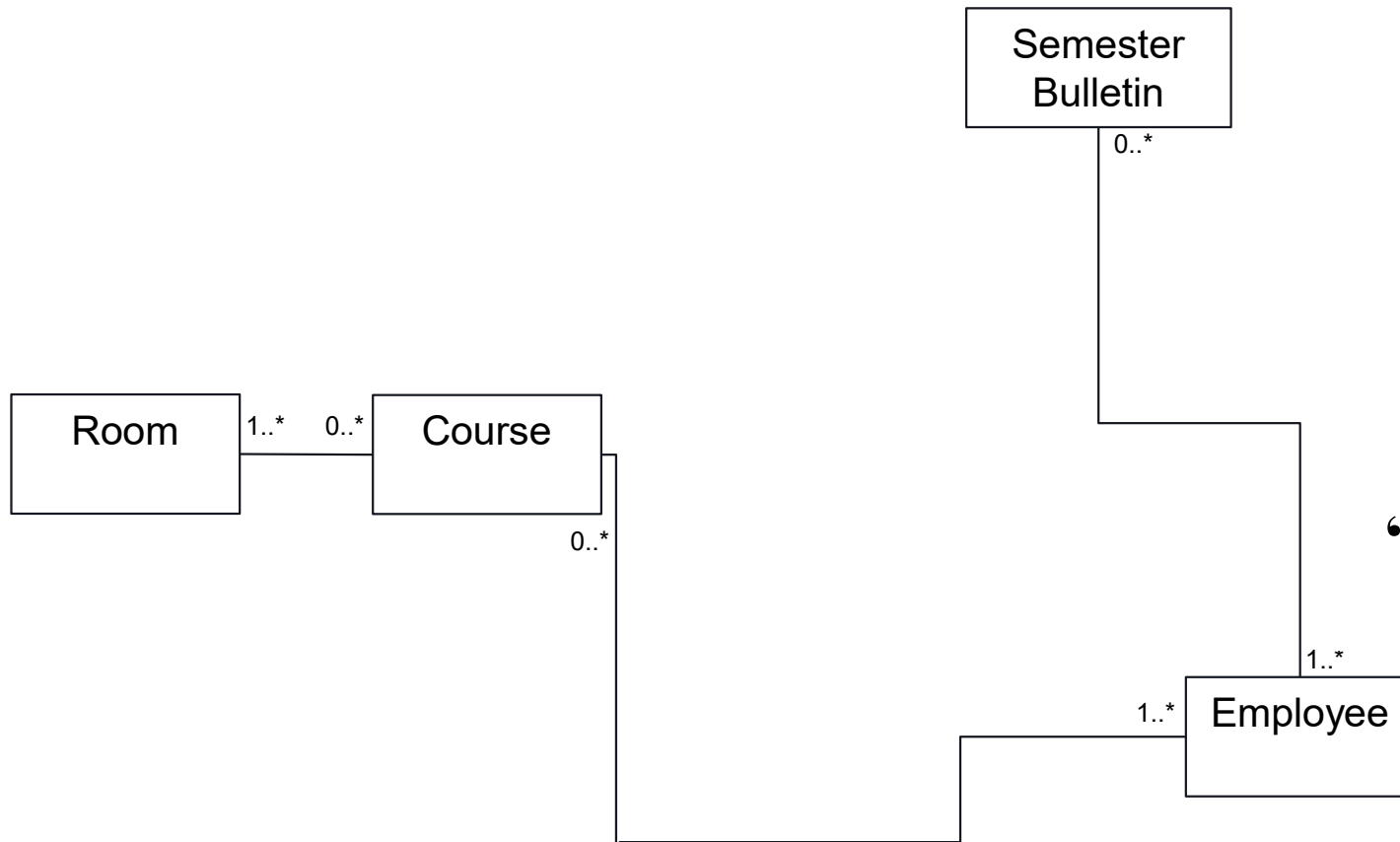
A general class (superclass) describes properties common to a group of specialized classes (subclasses)



Aggregation

Has-A

A superior object (the whole) consists of a number of inferior objects (parts)



Association “Just-Related”¹

A meaningful relation
between a number of
objects

¹Lecturer's home made term

Semester

Semester

Room

Course

Calendar

Coordination tools

Semester
Bulletin

Page
Resource

Document

File

Quiz

Persons

Person

Student

Employee

Teaching
Assistant

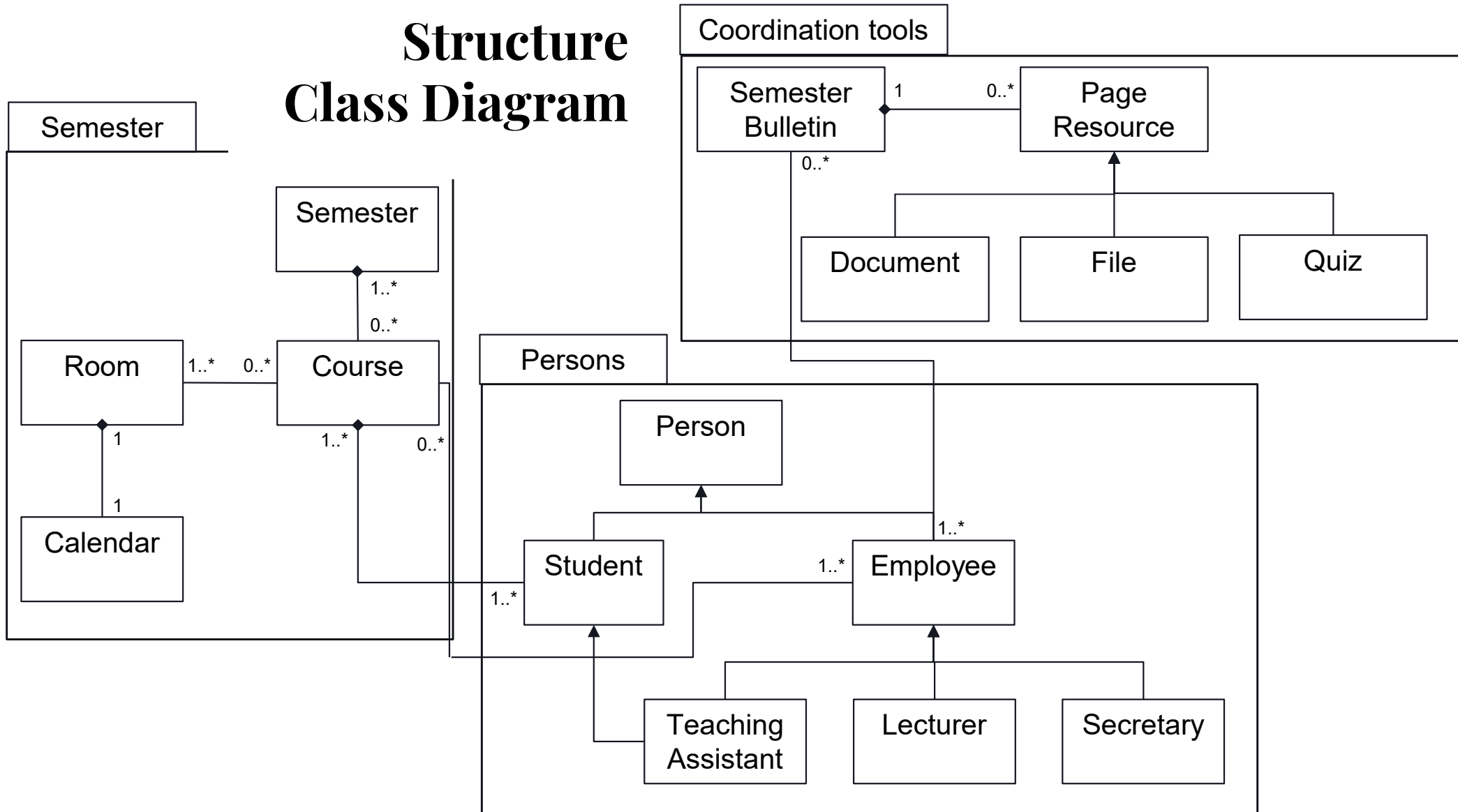
Lecturer

Secretary

Clusters

A collection of related
classes

Structure Class Diagram



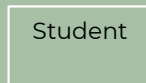
There are other relations!

E.g. "a Course Has-A Lecturer", "a Semester Has-A Semester Bulletin" etc... But the example is kept simple for pedagogical purposes.

Structure Through a Class Diagram

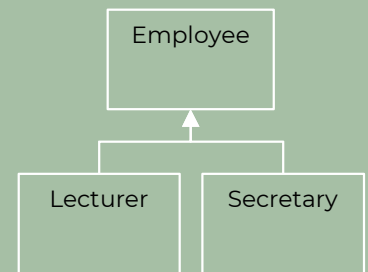
Class

A description of a collection of objects sharing structure, behavioural pattern and attributes.



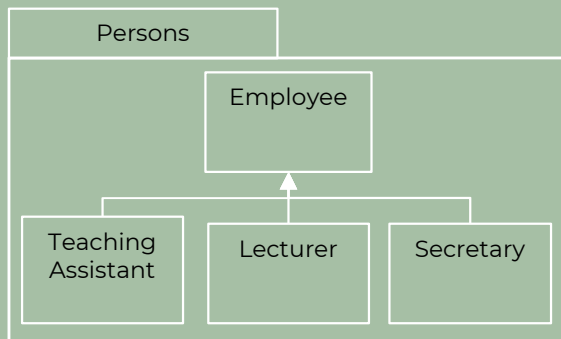
Generalization (Is-A)

A general class (superclass) describes properties common to a group of specialized classes (subclasses)



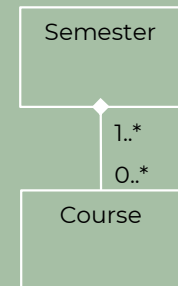
Cluster

A collection of related classes



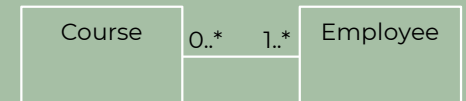
Aggregation (Has-A)

A superior object (the whole) consists of a number of inferior objects (parts)



Association ("Just-Related")

A meaningful relation between a number of objects. Not a defining property between objects.



Evaluation Criteria

Use Structures Correctly

E.g. don't mess up generalization and aggregation. Use the linguistic expressions Is-A, Has-A etc.

Conceptually True Structures

Names and structural relations must reflect users' understanding.

Structures Must Be Simple

Go for few classes and a few important relations. Avoid unnecessary generalizations and aggregations. Does the structure add relevance?

Uncovering Objects and Events

Exercise 2.1

Identify objects and events in the problem domain that are administrated, monitored or controlled in relation to F-klubbens bar.

For each identified object, argue why this belongs to the **problem domain**. Any objects that belongs to both the **problem domain** and the **application domain**? Argue why that is the case.

Make an **Event Table** to represent your findings and use the Affirmation Criteria to assess correctness of Classes and Events.

Structure through a Class Diagram

Exercise 2.2

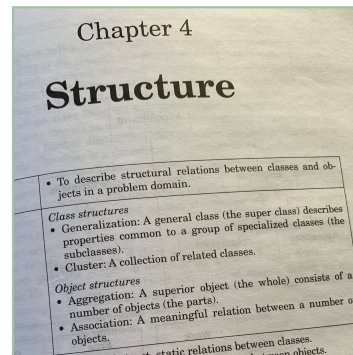
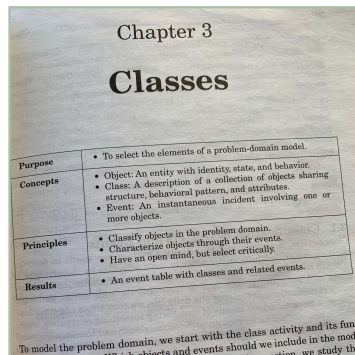
Identify how classes of objects are related for the F-klubben case in the previous assignment. Illustrate these relations in a consistent, unambiguous way of based on UML class diagram notation. Use the Evaluation Criteria to assess correctness of the structures.

Classes and Structure Activities for your projects

Exercise 2.3

Create an event table and a class diagram covering the problem domain in your semester projects. Use proper proper representations. Evaluate resulting event table and class diagram systematically.

Until we meet again



Supplementary material